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MSc Artificial Intelligence

ChestAI:

Multi-Disease Lung Pathology Detection from Chest X-Rays, Using Deep Learning, Transfer Learning, and Explainable AI with Clinical Image Enhancement Integration

Dataset: CheXpert v1.0-small Stanford University ML Group

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Abstract

Automated detection of thoracic pathologies from chest radiographs represents one of the most clinically impactful applications of deep learning in medical imaging. This paper presents ChestAI, a comprehensive end-to-end system for multi-label lung pathology detection from chest X-rays, developed as a final project for the MSc Computer Vision course at University Metropolitan Tirana (UMT). The system addresses the simultaneous detection of 14 distinct thoracic conditions using the CheXpert v1.0-small dataset (Irvin et al., 2019). Three CNN backbones were systematically evaluated - ResNet50, DenseNet121, and VGG16 - all employing an identical two-phase transfer learning strategy with class-weighted label-smoothed binary cross-entropy loss, enabling a fair and reproducible architectural comparison. VGG16 achieved the best mean AUC-ROC of 0.7492 on the held-out test set (N=5,000), outperforming both ResNet50 and DenseNet121 under controlled conditions. Model explainability was addressed through two gradient-based methods: Grad-CAM and Grad-CAM++, applied to VGG16's final convolutional layer (block5_conv3), with anatomically plausible activation maps produced for all 14 pathology classes.

A quantitative Pointing Game evaluation confirmed the spatial validity of Grad-CAM explanations, with positive mean activation deltas observed across the majority of pathology classes. A comprehensive fairness audit across sex and age subgroups revealed a sex-based demographic parity gap of 4.0% and an age-based equalized opportunity gap of 11.5%, with an intersectional gap of 15.2% between the best (Female, <40) and worst (Male, 40-60) subgroups.

A key novel contribution of this work is the Clinical X-Ray Enhancement Laboratory, an interactive preprocessing environment integrating 35 radiological image-processing algorithms - spanning spatial filtering, frequency-domain analysis, morphological operations, and contrast enhancement - with direct pipeline integration into VGG16 inference. This laboratory bridges classical Computer Vision theory with modern deep learning diagnostic practice, aligning with the UMT MSc curriculum across Topics 3 through 11. The complete system is deployed as a React and FastAPI web application offering real-time inference, Grad-CAM overlay visualization, interactive filter laboratory, and a demographic fairness dashboard. The full implementation comprises six Jupyter Notebooks covering EDA, preprocessing, three model training pipelines, explainability analysis, and fairness evaluation.

Keywords: chest X-ray; deep learning; multi-label classification; transfer learning; VGG16; DenseNet121; ResNet50; Grad-CAM; explainability; CheXpert; fairness; medical imaging AI; image preprocessing; convolutional neural networks; X-Ray Enhancement Laboratory; React; FastAPI

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1. Introduction

Chest radiography is among the most frequently performed diagnostic imaging procedures in clinical medicine worldwide. According to the World Health Organization, an estimated 2 billion chest X-rays are performed annually, and their interpretation by trained radiologists is a time-intensive and highly specialised task. The shortage of radiologists, particularly in low- and middle-income countries, combined with the increasing volume of imaging studies, has created a compelling need for automated, scalable, and accurate computer-aided detection systems.

The application of deep learning to chest X-ray analysis accelerated substantially following the publication of CheXNet (Rajpurkar et al., 2017), which demonstrated that a DenseNet121 trained on the ChestX-ray14 dataset could detect pneumonia at a level comparable to board-certified radiologists. This landmark result catalyzed a wave of subsequent work focused on multi-label pathology classification, model explainability, dataset curation, and fairness evaluation.

Despite significant progress, several challenges remain: (1) reliable simultaneous detection of 14 co-occurring thoracic pathologies; (2) handling of ambiguous uncertainty labels in radiologist annotations; (3) provision of interpretable explanations for model predictions; and (4) equitable performance across demographic subgroups. This paper addresses all four challenges within a unified system.

We present ChestAI, a comprehensive multi-label lung pathology detection system built on CheXpert v1.0-small. The system encompasses: a complete exploratory data analysis (EDA) pipeline (Notebook 1); a preprocessing and data augmentation framework (Notebook 2); a comparative evaluation of three CNN backbones - ResNet50, DenseNet121, VGG16 - under identical experimental conditions (Notebooks 3A, 3B); a model selection and comparison analysis (Notebook 4); Grad-CAM and Grad-CAM++ explainability with quantitative evaluation (Notebook 5); a demographic fairness analysis (Notebook 6); and an integrated Clinical X-Ray Enhancement Laboratory integrating 35 classical computer vision algorithms with direct VGG16 pipeline integration, deployed in a React and FastAPI web application.

1.1 Research Questions

This study addresses the following specific research questions:

- RQ1: What is the optimal CNN backbone architecture (ResNet50, DenseNet121, or VGG16) for multi-label chest pathology detection on CheXpert, under controlled and identical transfer learning conditions?
- RQ2: How effectively can Grad-CAM and Grad-CAM++ localize pathology-relevant regions in chest X-rays, and what quantitative evidence supports the spatial validity of these explanations?
- RQ3: Do demographic subgroups (biological sex and age) exhibit significantly different model performance, and what are the magnitudes and patterns of these disparities?
- RQ4: What role do classical image-processing filters (spatial, frequency-domain, morphological) play as preprocessing steps in the deep learning diagnostic pipeline for chest radiograph analysis?

1.2 Principal Contributions

The principal contributions of this work are:

1. A systematic, controlled comparison of three CNN backbones (ResNet50, DenseNet121, VGG16) under identical experimental conditions - same hyperparameters, augmentation strategy, loss function, and evaluation protocol - enabling a fair and reproducible architectural benchmark on CheXpert.
2. Implementation and quantitative evaluation of two gradient-based explainability methods (Grad-CAM and Grad-CAM++) on the best-performing model, including a Pointing Game analysis measuring the spatial quality of activation maps across all 14 pathology classes.

3. A comprehensive fairness audit across sex and age subgroups, computing demographic parity gap, equalized opportunity gap, and an intersectional subgroup analysis, providing a transparent quantification of the model's demographic performance disparities.
4. A novel Clinical X-Ray Enhancement Laboratory integrating 35 image-processing algorithms in five clinical categories with interactive parameter control, real-time clinical context annotations, filter chaining capability, and direct pipeline integration into VGG16 inference - bridging classical Computer Vision theory and clinical AI practice in a manner not previously described in the chest X-ray AI literature.
5. An end-to-end deployable web application (React + FastAPI) providing real-time VGG16 inference, Grad-CAM overlay visualization, demographic fairness dashboard, and the interactive filter laboratory - demonstrating full-stack ML engineering capability alongside the research contributions.

2. Literature Review

2.1 Deep Learning for Chest Radiograph Analysis

The systematic application of deep learning to chest X-ray interpretation began with Wang et al. (2017), who introduced ChestX-ray14 - a landmark dataset comprising 108,948 frontal-view radiographic images annotated with 14 disease labels derived automatically from radiology reports through natural language processing techniques. Prior to this work, the absence of large-scale, publicly available annotated chest X-ray corpora had severely constrained the development and rigorous evaluation of automated thoracic disease recognition systems. Wang et al. trained a DenseNet-121 architecture on this dataset and achieved AUC-ROC scores ranging from 0.745 for Hernia to 0.885 for Cardiomegaly, establishing the first comprehensive large-scale benchmark for automated multi-label thoracic disease classification and demonstrating that convolutional neural networks could learn diagnostically meaningful representations directly from radiographic data at population scale.

Rajpurkar et al. (2017) advanced the field significantly with CheXNet, demonstrating that a 121-layer DenseNet trained on ChestX-ray14 achieved an AUC-ROC of 0.768 for pneumonia detection - a performance level that exceeded the average diagnostic accuracy of four practising radiologists evaluated on the same held-out test set. This result constituted a watershed moment for clinical AI, offering the first compelling empirical evidence that deep learning systems could match or surpass human expert performance on a specific radiological classification task. Building on these foundations, Irvin et al. (2019) introduced CheXpert, the dataset employed in this study, which contains 224,316 chest radiographs sourced from 65,240 patients at Stanford University Medical Center. CheXpert provides expert annotations for 14 clinically relevant observations and introduces a defining ternary label system in which each finding is designated as positive (1.0), negative (0.0), or uncertain (-1.0). This uncertainty encoding, which affects approximately 17% of entries across all labels, represents one of CheXpert's most distinctive and methodologically challenging characteristics, as models must adopt explicit uncertainty-handling policies - such as ignoring uncertain labels, treating them as positive, or treating them as negative - each of which introduces different inductive biases and affects downstream performance in non-trivial ways.

Subsequent research has extended these foundational contributions along several complementary axes. Bustos et al. (2020) released PadChest, a large-scale Spanish-language radiograph dataset that expanded the label space to over 170 radiographic findings, probing model generalizability across linguistic and institutional boundaries. Guo et al. (2017) highlighted that high discriminative accuracy does not necessarily entail well-calibrated probability estimates, motivating post-hoc calibration methods such as temperature scaling that are particularly relevant in safety-critical clinical deployment. Azizi et al. (2021) demonstrated that self-supervised pretraining on large unlabelled medical imaging corpora can substantially improve downstream performance on limited labelled data, pointing toward more data-efficient training regimes. Zech et al. (2018) investigated multi-institutional generalization, revealing that models trained on data from a single hospital can exhibit significant performance degradation when applied to radiographs from different acquisition sites - a finding with direct implications for the external

validity of benchmark-trained systems. ChestAI situates itself within this trajectory by addressing model selection across three architectures (VGG16, ResNet50, DenseNet121), Grad-CAM explainability, demographic fairness analysis, and the integration of classical preprocessing pipelines - dimensions that, while each individually explored in the literature, remain underexplored in a unified, end-to-end framework targeting both diagnostic performance and responsible deployment criteria.

2.2 Transfer Learning and CNN Architectures

Transfer learning - the adaptation of models pre-trained on large general-purpose datasets (typically ImageNet, Russakovsky et al., 2015) to specialised medical imaging tasks - has become the de facto methodology in medical image analysis, driven primarily by the chronic scarcity of large, expertly annotated medical imaging datasets (Litjens et al., 2017). The underlying premise is that low- and mid-level visual features learned from millions of natural images - edges, textures, gradients, and compositional patterns - transfer meaningfully to radiographic domains, allowing models to converge faster and generalise better than training from random initialisation. This transfer is particularly valuable in chest X-ray analysis, where acquiring and labelling sufficient training data to train deep networks from scratch is both time-intensive and dependent on scarce radiologist expertise. Fine-tuning strategies vary across the literature, ranging from full network retraining to freezing early convolutional layers and updating only task-specific classification heads, with the optimal approach dependent on dataset size, domain shift magnitude, and architectural depth.

ResNet50 (He et al., 2016) introduced residual connections as a solution to the degradation problem observed in very deep networks, where increasing depth beyond a certain threshold paradoxically increased training error rather than reducing it. Each residual block computes the transformation $F(x) + x$, where the shortcut connection directly propagates the input to the output, ensuring unobstructed gradient flow during backpropagation and enabling stable training of networks with hundreds of layers. This architectural innovation fundamentally redefined the scalability of deep learning and established residual connectivity as a foundational design principle that has since influenced virtually every subsequent architecture. DenseNet121 (Huang et al., 2017) extended this connectivity paradigm further by proposing dense blocks in which every layer receives concatenated feature maps from all preceding layers within the block, rather than only from the immediately preceding layer. This dense connectivity maximises feature reuse, substantially reduces the number of parameters required to achieve competitive performance - approximately 7 million in the case of DenseNet121 - and encourages the learning of compact, non-redundant representations. It was precisely this architecture that Rajpurkar et al. (2017) employed in the original CheXNet system, and its relatively lightweight footprint makes it particularly attractive when computational resources or training data are limited. VGG16 (Simonyan and Zisserman, 2015), by contrast, adopts a philosophically simpler design philosophy: a strictly sequential stack of 3×3 convolutional layers organised into five progressively deeper blocks, followed by fully connected classification layers. Despite its architectural simplicity, VGG16 produces rich, hierarchical feature representations by incrementally building local pattern detectors of increasing receptive field and semantic complexity. Its parameter count of approximately 138 million is considerably larger than both ResNet50 and DenseNet121, which traditionally has been viewed as a disadvantage in terms of memory and computational cost, though this comes paired with substantial representational capacity in the convolutional backbone.

Under the controlled experimental protocol implemented in ChestAI, VGG16 demonstrated superior mean AUC-ROC performance relative to both ResNet50 and DenseNet121, despite - or perhaps because of - its larger parameter count and sequential convolutional structure. This result warrants careful interpretation: it suggests that VGG16's uniform, deeply stacked 3×3 filters may be particularly well-suited to extracting the texture- and structure-based discriminative features that characterise thoracic pathologies in grayscale radiographs, where subtle density gradients, interstitial patterns, and boundary irregularities are diagnostically significant. Residual and dense connectivity mechanisms, while highly effective in natural image recognition tasks that reward semantic abstraction, may introduce inductive biases less aligned with the low-contrast, high-noise characteristics of chest radiography. This architectural sensitivity to domain-specific feature statistics reinforces the importance of empirical model comparison rather than relying on benchmark rankings from natural image tasks when selecting architectures for medical imaging applications.

2.3 Explainable AI in Medical Imaging

The opacity of deep neural networks has motivated substantial research in explainable AI (XAI). Class Activation Maps (CAM, Zhou et al., 2016) were among the first methods to directly localise discriminative image regions. Grad-CAM (Selvaraju et al., 2020) generalised CAM to any CNN architecture using gradients flowing into the final convolutional layer to compute class-discriminative importance weights, requiring no architectural modification. Grad-CAM++ (Chattopadhyay et al., 2018) improved upon Grad-CAM by incorporating second-order gradient information, yielding better spatial coverage for multi-instance activations - particularly relevant for chest X-rays where pathology may manifest at multiple anatomical loci simultaneously.

Quantitative evaluation of XAI methods in medical imaging remains an open problem. This study employs the Pointing Game paradigm (Zhang et al., 2018), comparing the maximum Grad-CAM activation between positive and negative samples as a proxy for localization fidelity - a method that requires no bounding box annotations and is therefore applicable to the full CheXpert test set.

2.4 Algorithmic Fairness in Medical AI

The fairness of AI systems in healthcare has emerged as a critical and rapidly expanding research area, motivated by mounting empirical evidence that algorithmic decision-support tools can encode, amplify, and perpetuate the same structural inequities present in the historical data on which they are trained. The consequences in clinical contexts are particularly grave: biased predictions do not merely represent abstract statistical disparities but translate directly into differential quality of care, misallocation of medical resources, and systematically worse health outcomes for already disadvantaged populations. Obermeyer et al. (2019) provided one of the most influential demonstrations of this dynamic, showing that a widely deployed commercial risk-stratification algorithm assigned systematically lower risk scores to Black patients relative to White patients with equivalent illness severity, effectively denying Black patients access to high-complexity care management programs at scale. The root cause was an apparently neutral design choice - using healthcare expenditure as a proxy for health need - which failed to account for the fact that structural barriers to healthcare access meant that Black patients historically generated lower costs despite comparable or greater medical need. This study fundamentally reframed algorithmic fairness not as a theoretical concern but as an active and measurable harm operating within deployed clinical systems.

In the specific domain of chest X-ray AI, Seyyed-Kalantari et al. (2021) conducted a systematic fairness audit across five large chest radiograph datasets - including CheXpert, the dataset used in this study - and documented consistent underperformance for female patients, patients aged below 40, and patients from certain racial groups, even when overall model performance metrics appeared satisfactory. Critically, this underperformance was observed across multiple model architectures and training regimes, suggesting that the disparities were driven primarily by imbalances in dataset composition and label distributions rather than by specific modelling choices. Larson et al. (2021) further quantified these disparities at the pathology level, documenting AUC-ROC differences of 3–8 percentage points between male and female patients for specific conditions, with the magnitude and direction of the gap varying across diseases - a finding that underscores the inadequacy of reporting aggregate performance metrics alone and the necessity of disaggregated evaluation as a standard component of model validation in medical AI.

The fairness analytical framework employed in ChestAI follows the formal criteria established by Hardt et al. (2016), who provided rigorous mathematical definitions of Demographic Parity - requiring that positive prediction rates be equal across demographic groups regardless of ground-truth label - and Equalized Opportunity - a relaxation that requires equal true positive rates across groups, acknowledging that differential base rates may make strict demographic parity clinically inappropriate in some settings. These two criteria capture complementary aspects of fairness: the former concerns distributional equity in predictions, while the latter concerns equity in the model's utility for patients who genuinely have a condition. ChestAI extends this dyadic framework to intersectional analysis following Kearns et al. (2018), who demonstrated formally that models appearing fair along individual demographic axes - such as sex or age considered independently - may nonetheless exhibit substantial disparities for specific subgroup intersections, such as young female patients or elderly patients from underrepresented racial groups. This intersectional lens is particularly important in chest radiography, where pathology

prevalence, imaging characteristics, and clinical presentation can differ simultaneously across multiple demographic dimensions, making single-axis fairness evaluation an insufficient safeguard against real-world discriminatory outcomes.

2.5 Classical Image Processing in Deep Learning Pipelines

Classical computer vision filtering techniques - predating deep learning by several decades - remain practically indispensable as preprocessing steps in medical imaging AI pipelines, not as legacy holdovers but as principled signal processing operations that address fundamental properties of radiographic image formation that neural networks alone cannot be guaranteed to handle optimally. Raw chest radiographs present a constellation of challenges that classical methods are specifically designed to mitigate: non-uniform exposure across anatomical regions, low inherent contrast between pathological and healthy tissue, acquisition noise from detector electronics and photon statistics, and the superimposition of overlapping anatomical structures onto a two-dimensional projection plane. Addressing these challenges prior to feature extraction can substantially improve the diagnostic signal available to downstream deep learning models, particularly for subtle findings whose radiographic signatures occupy narrow contrast ranges or fine spatial frequency bands.

Contrast Limited Adaptive Histogram Equalization (CLAHE, Pizer et al., 1987) represents the clinical standard for enhancing local contrast in chest radiographs and has seen widespread adoption precisely because it addresses a fundamental limitation of global histogram equalization: the inability to adapt contrast enhancement to spatially varying local image statistics. By operating on small overlapping tile regions independently and applying a clip limit that caps the local contrast amplification factor, CLAHE prevents the noise amplification in homogeneous tissue regions that renders global equalization clinically counterproductive while simultaneously revealing infiltrative patterns - ground-glass opacities, interstitial edema, early consolidation, and subtle reticular markings - that are frequently invisible or ambiguous in raw radiographs acquired under suboptimal exposure conditions. This property is of particular clinical significance in the context of the CheXpert label set, where pathologies such as Edema and Lung Opacity manifest as precisely the kind of low-contrast diffuse infiltrative changes that CLAHE is designed to surface.

Edge detection operators constitute a complementary family of classical preprocessing tools that enhance the representation of anatomically meaningful boundaries by computing local intensity gradient magnitudes and orientations. The Sobel operator (Duda and Hart, 1973) applies directional derivative kernels to detect horizontal and vertical gradient components, providing a computationally efficient approximation to first-order edge structure that is robust to moderate noise. The Laplacian and Laplacian of Gaussian (LoG, Marr and Hildreth, 1980) detect regions of rapid intensity change via second-order derivatives, with the Gaussian smoothing component in LoG providing essential noise suppression before differentiation amplifies high-frequency artifacts. The Canny edge detector (Canny, 1986) extends this framework with non-maximum suppression and hysteresis thresholding, producing thin, well-localised edge maps that accurately delineate tissue interfaces, vessel walls, pleural margins, and nodule contours - structural boundaries whose integrity is diagnostically significant across multiple CheXpert pathology categories including Pneumothorax, Pleural Effusion, and Cardiomegaly. Frequency-domain filtering via the Fast Fourier Transform provides yet another analytical dimension by enabling targeted suppression or enhancement of specific spatial frequency bands independently of their spatial location. Low-pass filtering attenuates high-frequency noise while preserving the gross anatomical structures that define cardiac silhouette, diaphragmatic contour, and mediastinal width, whereas high-pass filtering emphasises fine structural details such as nodule edges, vascular markings, and interstitial line patterns that carry diagnostic weight for pulmonary pathologies. Morphological operations - erosion, dilation, opening, closing, and top-hat transforms - complement these frequency and gradient-based approaches by operating on the geometric structure of image regions rather than on intensity values directly, making them fundamental tools for lung region segmentation, rib suppression, and the removal of acquisition artifacts and overlaid text annotations that would otherwise confound deep learning feature extraction (Serra, 1982; Shen et al., 2015).

Despite this well-established and extensively documented value, a significant and largely unaddressed gap exists in the design of published chest X-ray AI systems: the overwhelming majority treat preprocessing as a fixed, opaque pipeline step that is invisible to end users, clinicians, and researchers

who interact with the deployed system. Preprocessing choices - which filters are applied, in what sequence, with what parameter settings - exert non-trivial influence on model inputs and therefore on predictions, yet these choices are rarely exposed, documented, or made explorable in existing clinical AI interfaces. ChestAI directly addresses this transparency deficit through the Clinical X-Ray Enhancement Laboratory, a dedicated interactive module that makes the full classical computer vision preprocessing layer visible, configurable, and directly integrated with the deep learning inference system. By enabling clinicians and researchers to apply, adjust, and combine preprocessing operations in real time and observe their downstream effect on model predictions, ChestAI transforms preprocessing from an opaque technical implementation detail into an explicit, auditable, and pedagogically valuable component of the diagnostic workflow.

3. Dataset and Exploratory Data Analysis

3.1 CheXpert Dataset

This study uses the CheXpert v1.0-small dataset (Irvin et al., 2019), produced by the Stanford University Machine Learning Group. CheXpert is one of the largest publicly available chest radiograph datasets annotated for a comprehensive set of thoracic pathologies and was specifically selected for its ternary label system, which reflects the genuine ambiguity present in radiological interpretation.

Property	Value
Total training studies	223,414
Validation studies (gold standard)	234 (radiologist-annotated by 5 board-certified radiologists)
Number of patients	~65,240
Number of pathology labels	14 (multi-label, non-exclusive)
View types included	Frontal AP/PA (191,027) + Lateral (32,387)
Image format	JPEG grayscale, variable resolution
Label types	Positive (1.0), Negative (0.0), Uncertain (-1.0), NaN
Dataset source	Stanford Hospital radiology archive
Dataset DOI	https://doi.org/10.71718/y7pj-4v93

Table 3.1 - CheXpert v1.0-small Dataset Summary

3.2 Label System and Uncertainty Handling

A defining characteristic of CheXpert is its ternary label system, reflecting the inherent uncertainty in radiological annotation. Across all 14 pathology labels, approximately 17% of entries carry the uncertain value (-1.0), with particularly high uncertainty rates (>30%) for Edema, Atelectasis, Pleural Effusion, and Cardiomegaly - precisely the four conditions where clinical consequences of missed detection are most severe.

Label Value	Meaning	Description
1.0	Positive	Pathology explicitly confirmed as present
0.0	Negative	Pathology explicitly confirmed as absent
-1.0	Uncertain	Radiologist expressed uncertainty about presence
NaN	Not mentioned	Label not referenced in the radiology report

Table 3.2 - CheXpert Ternary Label System

3.3 Exploratory Data Analysis - Key Findings (Notebook 1)

Notebook 1 conducted a comprehensive EDA across four analytical dimensions: class distribution, patient demographics, image resolution, and label co-occurrence. Taken together, these analyses reveal a dataset whose statistical properties introduce non-trivial modelling challenges - severe label imbalance, demographic skew, resolution heterogeneity, and structured label dependencies - each of which required explicit methodological responses in the preprocessing and training pipeline. The findings directly shaped all subsequent design decisions from uncertainty label handling to augmentation strategy to loss function formulation.

3.3.1 Class Imbalance

Severe class imbalance was observed across the 14 pathology labels, spanning more than an order of magnitude in positive label frequency between the most and least prevalent conditions. Support Devices (75,078 positives) and Pleural Effusion (61,529 positives) dominate the positive label distribution, reflecting the high prevalence of intensive care and post-procedural radiographs in the Stanford Hospital archive from which CheXpert was derived. At the opposite extreme, Pneumonia (4,576 positives), Lung Lesion (5,892 positives), and Pleural Other (5,936 positives) are rare in absolute terms, with positive-to-total ratios below 3% - meaning that for every positive training example of Pneumonia, there are more than 33 negative examples. For the rarest labels the imbalance ratio exceeds 40:1, creating a strong prior toward the negative class that an unconstrained binary cross-entropy loss function would exploit trivially: a model that predicts negative for all cases on these labels would achieve over 97% binary accuracy while providing no diagnostic information whatsoever.

This imbalance has direct implications for both model training and clinical utility. During training, gradient updates from the majority negative class would dominate parameter updates under a standard unweighted loss, causing the model to converge to a high-accuracy but clinically useless negative-predicting regime for rare labels. The mitigation adopted in this work - per-label class-weighted loss functions, with weights computed as the inverse frequency of the positive class for each label independently - rebalances gradient contributions such that each positive training example for a rare label contributes proportionally more to parameter updates than a negative example, incentivising the model to learn discriminative representations for all 14 classes rather than collapsing to the majority class for the rarest. The severity of the imbalance observed in the EDA directly informed the decision to compute per-label weights independently rather than applying a single global class weight, acknowledging that the imbalance ratio varies substantially across labels and that a single weight would simultaneously over-correct for common labels and under-correct for rare ones.

3.3.2 Patient Demographics

The dataset exhibits a pronounced age skew toward older patients, with a median age exceeding 55 years - a distribution that reflects the higher prevalence of thoracic pathologies in elderly populations and the clinical context of a tertiary referral hospital, where radiographs are ordered preferentially for patients with established or suspected cardiorespiratory disease. The age distribution is right-skewed with a long tail toward the very elderly, and patients below 40 years constitute a small minority of the training set. The sex distribution is moderately imbalanced, with approximately 56% male and 44% female patients in the frontal training set, consistent with the slightly higher prevalence of several thoracic conditions - including ischemic heart disease and lung cancer - in male populations.

These demographic imbalances are not merely descriptive statistics: they constitute a direct mechanistic explanation for the performance disparities documented in the fairness analysis of Section 6. A model trained on a dataset where elderly and male patients are disproportionately represented will receive more gradient updates from these majority subgroups and will develop feature representations that are better calibrated for their radiographic presentations. Younger patients and female patients, whose pathological presentations may differ in distribution, morphology, and co-occurrence pattern from the majority

training population, will be systematically underrepresented in the gradient signal, producing feature representations that generalise less reliably to these subgroups. The age-based equalized opportunity gap of 11.5% and the sex-based demographic parity gap of 4.0% documented in Notebook 6 are thus predictable consequences of the demographic composition revealed in this EDA, rather than unexpected post-hoc findings - a causal chain from dataset structure to model behaviour that the EDA makes explicit before any model is trained.

3.3.3 Image Resolution Variability

Analysis of 300 randomly sampled frontal images revealed substantial resolution variability across the CheXpert training set, with image heights ranging from 320 to 3,048 pixels and widths from 320 to 3,056 pixels. This variability reflects the heterogeneous acquisition history of the Stanford Hospital archive, which spans multiple scanner generations, detector technologies, and acquisition protocols accumulated over years of clinical practice. Portable AP radiographs - common in critically ill patients who cannot be transported to the radiology department - are typically acquired at lower resolution than departmental PA radiographs, introducing a systematic correlation between acquisition type, patient acuity, and image resolution that could confound resolution-based preprocessing decisions.

All images were standardized to 224×224 pixels during preprocessing to satisfy the fixed input dimensionality requirements of the ImageNet-pretrained VGG16, ResNet50, and DenseNet121 backbones. For the majority of images - which were acquired at resolutions substantially exceeding 224 pixels in both dimensions - this standardization introduces significant downsampling, with bilinear interpolation compressing the original high-resolution radiographic detail into a 50-fold smaller pixel grid for images acquired at the maximum observed resolution of approximately 3,000×3,000 pixels. The clinical consequences of this downsampling are not uniform across pathologies: structural findings with large-scale radiographic signatures (cardiomegaly, large pleural effusion, major consolidation) are largely preserved at 224×224 resolution, while fine-detail findings such as subtle interstitial markings, small nodules, early ground-glass changes, and fine reticular patterns may be substantially degraded or lost. This resolution constraint represents a fundamental trade-off between computational tractability and diagnostic information preservation, and partially accounts for the lower per-label AUC observed for pathologies whose discriminative radiographic features operate at fine spatial scales.

3.3.4 View Type Distribution

Of the 223,414 training studies, 191,027 (85.5%) are frontal-view radiographs and 32,387 (14.5%) are lateral-view radiographs, yielding a substantial majority of frontal acquisitions that reflects both the clinical preference for the frontal projection as the primary diagnostic view and the higher proportion of portable AP frontal radiographs in the inpatient population. Following the established practice of published CheXpert benchmarks (Irvin et al., 2019; Rajpurkar et al., 2017), only frontal-view images were retained for training and evaluation in this study. This decision is justified on two grounds: first, the frontal projection provides a consistent anatomical perspective from which all 14 CheXpert pathology labels can be assessed, whereas lateral views have differential diagnostic value for specific conditions (pleural effusion, posterior consolidation) but limited value for others (cardiomegaly, pneumothorax in most presentations); second, mixing frontal and lateral views without explicit view-conditioning would introduce a confounding source of anatomical variability - mediastinal structures appear entirely differently between projections - that a single model trained without projection-type input would need to learn to disentangle implicitly, adding unnecessary complexity and a potential source of misclassification.

3.3.5 Label Co-occurrence Analysis

Multi-label analysis using the phi coefficient as a measure of pairwise label association revealed frequent and clinically meaningful co-occurrence patterns among the 14 pathology labels, confirming that the multi-label formulation is not merely a modelling convenience but a clinical necessity imposed by the genuine co-occurrence structure of thoracic pathology in a hospital population. The three highest pairwise co-occurrence coefficients were: Pleural Effusion and Cardiomegaly ($\Phi = 0.41$), Edema and Pleural Effusion ($\Phi = 0.38$), and Support Devices and Pleural Effusion ($\Phi = 0.35$).

These associations are not statistical artifacts but reflect well-established pathophysiological relationships. Cardiomegaly and Pleural Effusion co-occur at high rates because cardiac enlargement, most commonly from congestive heart failure, directly precipitates pleural fluid accumulation through elevated pulmonary venous pressure and impaired lymphatic drainage - the same underlying mechanism that produces Edema, explaining the near-equally strong Edema–Pleural Effusion association. Support Devices and Pleural Effusion co-occur because patients requiring central venous catheters, endotracheal tubes, or chest drains - the primary Support Device categories in CheXpert - are disproportionately critically ill patients with cardiorespiratory compromise, for whom pleural effusion is both a common finding and a frequent indication for device placement. These co-occurrence patterns have practical implications for model training and evaluation: a model that has learned the Cardiomegaly–Pleural Effusion association may achieve inflated AUC for Pleural Effusion by leveraging cardiac silhouette features as a proxy, rather than directly detecting the pleural fluid signal. Distinguishing genuine pathology detection from co-occurrence-driven label inference is not possible with the available data but represents an important consideration in interpreting per-label AUC results, particularly for label pairs with Φ coefficients exceeding 0.35.

4. Methodology

4.1 Data Preprocessing Pipeline (Notebook 2)

4.1.1 Uncertainty Label Strategy: U-Mixed

Based on EDA findings and the recommendations of Irvin et al. (2019), a U-Mixed uncertainty handling strategy was adopted. This strategy applies U-Ones (map $-1.0 \rightarrow 1.0$) for four clinically critical labels where missed detections are most costly: Edema, Atelectasis, Pleural Effusion, and Cardiomegaly. For all remaining labels, U-Zeros (map $-1.0 \rightarrow 0.0$) is applied, treating uncertain cases conservatively as negative. This hybrid strategy reflects the clinical principle of maximizing sensitivity for high-severity, treatable conditions while maintaining specificity for less critical findings.

4.1.2 Image Preprocessing Pipeline

All images underwent the following standardized preprocessing sequence, implemented using TensorFlow's tf.data API for efficient parallel loading and GPU-accelerated execution:

6. Read as JPEG grayscale (CheXpert images are single-channel).
7. Bilinear resize to 224×224 pixels - the standard ImageNet input dimension.
8. Replicate the single grayscale channel to 3 channels (pseudo-RGB), required to match the input expectations of ImageNet-pretrained models.
9. Cast to float32 and normalize pixel values from [0, 255] to [0.0, 1.0].

4.1.3 Data Augmentation Strategy

Training-set augmentation was applied to improve generalization and reduce overfitting. All augmentation parameters were chosen to reflect plausible acquisition variability in real-world clinical chest radiography - augmentations that a patient might realistically present, and that do not alter the clinical interpretation of the image.

Augmentation	Parameters	Clinical Rationale
Horizontal flip	$p = 0.50$	Left-right symmetry in chest anatomy; situs inversus is sufficiently rare
Random rotation	$\pm 5^\circ$ (0.05 rad)	Patient positioning variability during acquisition
Random zoom	$\pm 8\%$ (-0.08 to $+0.08$)	Distance and magnification variability between acquisition systems
Random contrast	factor = 0.10	X-ray exposure and detector sensitivity variability

Table 4.1 - Data Augmentation Strategy with Clinical Rationale

4.1.4 Dataset Split

The 191,027 frontal training studies were stratified split - stratified on the Pleural Effusion label to ensure balanced representation across splits.

Split	N Samples	Proportion	Purpose
Training	50,000	80%	Model parameter optimisation via gradient descent
Validation (internal)	5,000	10%	Hyperparameter tuning and early stopping
Test (held-out)	5,000	10%	Final unbiased evaluation - not touched during training
Gold standard (official)	202	-	CheXpert official validation (radiologist-annotated)

Table 4.2 - Dataset Split Configuration

Note: The training subset of 50,000 images was used due to computational constraints on the Kaggle P100 GPU environment; the full frontal training set comprises approximately 190,000 studies.

4.1.5 Class Weighting

Weighted binary cross-entropy addressed class imbalance. Per-label positive weights were computed as $w_j = (N - P_j) / P_j$, where N is the total number of training samples and P_j is the number of positive samples for label j . These weights penalize false negatives on rare labels more heavily, ensuring the model does not collapse to predicting the majority negative class for rare pathologies such as Pneumonia or Lung Lesion.

4.2 Model Architectures

Three CNN backbone architectures were evaluated in a controlled experiment. All models shared an identical classification head and training configuration; the only variable was the backbone architecture, ensuring fair comparison.

4.2.1 Shared Classification Head

Layer	Purpose
Backbone (frozen, Phase 1)	Feature extraction from chest radiograph
GlobalAveragePooling2D	Spatial dimension reduction; CAM-compatible
Dense(512)	High-level feature combination
BatchNormalization	Training stabilisation; gradient flow
Dropout(0.5)	Regularization against overfitting
Dense(256)	Further feature compression
BatchNormalization	Training stabilisation
Dropout(0.3)	Regularization
Dense(14, sigmoid)	Multi-label probability output for 14 pathologies

Table 4.3 - Shared Classification Head Architecture

4.2.2 ResNet50

ResNet50 (He et al., 2016) introduces residual connections computing $F(x) + x$, enabling stable training of very deep networks by addressing the vanishing gradient problem. ResNet50 comprises 50 weight layers with approximately 25 million parameters, organized in four residual stages. In Phase 2 fine-tuning, the last 30 layers (Stage 5, conv5_block*) were unfrozen, allowing the final residual stage to adapt to chest radiograph features.

4.2.3 DenseNet121

DenseNet121 (Huang et al., 2017) introduces dense connectivity: each layer l receives as input the concatenation of feature maps from all preceding layers $[h_0, h_1, \dots, h_{l-1}]$. This design maximises feature reuse and gradient flow with approximately 7 million parameters - the most parameter-efficient architecture evaluated. DenseNet121 is the architecture of the original CheXNet system (Rajpurkar et al., 2017). In Phase 2 fine-tuning, the last 80 layers (dense_block4 and transition layers) were unfrozen.

4.2.4 VGG16

VGG16 (Simonyan and Zisserman, 2015) employs a uniform architecture of sequential 3×3 convolutional layers arranged in 5 blocks (block1 through block5), followed by global average pooling. Despite its larger parameter count (~138 million), VGG16 demonstrated superior performance on this task, with the deep stacking of small convolutional filters producing rich hierarchical feature representations capturing both fine-grained textures and coarser structural patterns. In Phase 2 fine-tuning, blocks 4 and 5 were unfrozen. The last convolutional layer - block5_conv3 (spatial resolution 14×14 for 224×224 input) - serves as the target layer for Grad-CAM computation.

4.3 Training Protocol

An identical two-phase transfer learning protocol was applied to all three architectures, ensuring a fair comparison across backbones.

Phase 1 - Feature Extraction (10 Epochs)

Backbone weights are frozen; only the classification head is trained. Initial learning rate: 1×10^{-3} (Adam optimizer). This phase allows the randomly-initialized head weights to converge before exposing the backbone to gradients, preventing catastrophic interference with pretrained features.

Phase 2 - Fine-Tuning (up to 20 Epochs)

The final layers of the backbone are unfrozen (architecture-specific, see Sections 4.2.2–4.2.4). Learning rate: 1×10^{-5} with cosine annealing (Loshchilov and Hutter, 2017). Fine-tuning at a lower learning rate adapts pretrained feature representations to the chest X-ray domain without catastrophically forgetting ImageNet features - a critical balance for small-data medical imaging.

4.3.1 Hyperparameters and Loss Function

Hyperparameter	Value	Rationale
Batch size	32	Memory-efficient for P100 GPU; stable gradient estimates
Phase 1 learning rate	1×10^{-3}	Standard for head training with frozen backbone
Phase 2 learning rate	1×10^{-5}	Conservative fine-tuning to preserve ImageNet features
LR schedule	Cosine annealing	Smooth convergence; avoids sharp LR transitions
Early stopping patience	8 epochs (val_auc)	Prevent overfitting; monitored on validation AUC
ReduceLROnPlateau factor	0.3 (patience = 3)	Adaptive LR reduction on plateaus
Label smoothing ϵ	0.05	Prevent overconfidence; improve calibration
Loss function	Weighted BCE + Label Smoothing	Addresses class imbalance + calibration
Optimizer	Adam	Adaptive learning rates; robust to gradient noise
Primary metric	AUC-ROC (macro-averaged)	Threshold-free; appropriate for imbalanced multi-label
Random seed	42 (all experiments)	Reproducibility across all three backbone experiments

Table 4.4 - Hyperparameters and Rationale

4.4 Grad-CAM and Grad-CAM++ Explainability (Notebook 5)

4.4.1 Grad-CAM Implementation

Grad-CAM (Selvaraju et al., 2020) computes a class-discriminative localization map for a target class c by weighting each feature map in the final convolutional layer by the global-average-pooled gradient of the class score with respect to that feature map, then applying a ReLU nonlinearity to retain only the spatially positive contributions to the prediction. Formally, the localization map is defined as:

$$L^c = \text{ReLU} \left(\sum_k \alpha_k^c \cdot A^k \right)$$

where A^k denotes the k -th feature map of the target convolutional layer and the neuron importance weight α_k^c is computed as:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

with Z denoting the total number of spatial locations in the feature map and $\partial y^c / \partial A_{ij}^k$ representing the partial derivative of the class score y^c with respect to the activation at spatial position (i, j) of the k -th feature map. The global average pooling over all spatial positions compresses the full gradient tensor into a scalar importance weight per feature map channel, capturing the aggregate contribution of each channel to the class score while discarding spatial gradient structure - a design choice that produces smooth, coarse localization maps appropriate for identifying broad regions of discriminative activation rather than pixel-precise boundaries.

In this implementation, Grad-CAM was applied to the block5_conv3 layer of VGG16, which constitutes the final convolutional layer of the network before the global average pooling and fully connected classification head. At an input resolution of 224×224 pixels, block5_conv3 produces feature maps of spatial resolution 14×14, encoding high-level semantic features with a receptive field that spans a substantial portion of the input image. This choice of target layer reflects the standard Grad-CAM practice of targeting the deepest convolutional layer, where feature representations are maximally class-discriminative while retaining sufficient spatial resolution to produce interpretable localization maps. The resulting 14×14 activation maps were resized to 224×224 via bilinear interpolation to match the input image dimensions, and the upsampled heatmap was overlaid on the original grayscale radiograph using the COLORMAP_JET colormap at a blending factor of $\alpha = 0.45$ - a value empirically selected to balance heatmap visibility with preservation of underlying anatomical detail, ensuring that the radiographic structures informing the activation pattern remain visible to clinical reviewers. Gradient computation was implemented using TensorFlow's `tf.GradientTape` in inference-only mode, with the model's trainable weights frozen during the gradient computation pass, ensuring that Grad-CAM visualization introduces no training overhead and can be computed at inference time for any input image without modifying model state.

4.4.2 Grad-CAM++ Implementation

Grad-CAM++ (Chattopadhyay et al., 2018) extends the Grad-CAM formulation by replacing the global-average-pooled gradient weights with spatially resolved, pixel-wise weighting coefficients that more accurately capture the relative importance of each spatial location in the feature map. The fundamental limitation that Grad-CAM++ addresses is the information loss inherent in global average pooling of gradients: when a pathology appears at multiple anatomical loci simultaneously - bilateral pleural effusion occupying both costophrenic angles, diffuse interstitial edema distributed across both lung fields, multifocal pneumonia involving non-contiguous segments - the global average pooling operation in Grad-CAM averages gradient contributions from all spatial positions, producing a weight that reflects the mean discriminative contribution rather than the local contribution of each instance. This averaging can cause Grad-CAM to collapse to the single strongest instance of the pathology while attenuating the signal from secondary instances, yielding activation maps that incompletely represent the spatial extent of the finding.

Grad-CAM++ resolves this by deriving pixel-wise weighting coefficients α^{kc}_{ij} using second-order partial derivatives of the class score with respect to the feature map activations. These higher-order gradient terms encode not merely the direction and magnitude of each activation's contribution to the class score, but also the curvature of the class score function with respect to each spatial location - information that allows the weighting scheme to assign higher importance to spatial positions that are locally discriminative even when their global gradient contribution is diluted by averaging across a large feature map containing multiple pathological instances. The resulting localization maps exhibit broader, more spatially distributed activation patterns compared to Grad-CAM for multi-instance pathologies, at the cost of somewhat reduced sharpness for focal, single-instance findings where the global average pooling of standard Grad-CAM is already adequate. In the ChestAI implementation, Grad-CAM++ was applied to the same block5_conv3 target layer as Grad-CAM, using the same bilinear upsampling and COLORMAP_JET overlay procedure, ensuring that differences in the resulting heatmaps are attributable entirely to the weighting scheme rather than to any difference in target layer, upsampling method, or visualization parameters. Both methods are computed on demand for any selected pathology class through the ChestAI web application interface, enabling side-by-side comparison.

4.4.3 Pointing Game Evaluation

The Pointing Game (Zhang et al., 2018) was employed as the primary quantitative evaluation framework for assessing the spatial validity of Grad-CAM explanations across the full 14-label CheXpert pathology set, chosen specifically because it requires no bounding box annotations or expert-provided localization ground truth - prerequisites that CheXpert does not satisfy for the majority of its labels and that would render annotation-dependent localization metrics inapplicable at scale.

The evaluation protocol is as follows. For each pathology class independently, 200 confirmed positive samples (binary label = 1.0 after U-Mixed uncertainty mapping) and 200 confirmed negative samples (binary label = 0.0) were drawn from the held-out test set, stratified to ensure balanced representation across the prediction probability distribution rather than sampling exclusively from high-confidence predictions. For each sample, a Grad-CAM heatmap was computed for the target pathology class and the maximum activation value within the resized

224×224 heatmap was recorded as the pointing game score for that sample. The Pointing Game delta is then defined as:

$$\Delta = \text{mean_max_activation}(\text{positive samples}) - \text{mean_max_activation}(\text{negative samples})$$

This metric operationalizes a specific and testable hypothesis about spatial validity: if Grad-CAM explanations are spatially coherent with pathological features, then the maximum activation within the heatmap should be systematically higher when the target pathology is genuinely present than when it is absent, because the model should attend more strongly to pathology-specific regions in positive cases. The metric is threshold-free - it does not require a binary decision about whether the maximum activation falls within a defined ground-truth bounding box, as in the original Pointing Game formulation of Zhang et al. (2018) - making it applicable without localization annotations while retaining sensitivity to systematic differences in spatial attention between positive and negative cases. A positive Δ across a balanced sample of 400 cases per label constitutes evidence that the model's spatial attention is not random with respect to pathology presence, providing a principled lower bound on spatial validity that is both annotation-free and statistically robust to individual outlier cases. The choice of N=200 per class per group was determined by the availability of confirmed positive samples for the rarest pathology labels in the test set, ensuring that the evaluation is conducted on a consistent balanced protocol across all 14 classes rather than using variable sample sizes that would make cross-label delta comparisons unreliable.

4.5 Fairness Analysis Methodology (Notebook 6)

Fairness was evaluated using the CheXpert official validation set (valid.csv), which provides demographic metadata (Sex, Age) linked to each study. Three fairness dimensions were assessed:

- Sex-based analysis: AUC-ROC computed separately for Male and Female patients across all 14 pathology labels. The demographic parity gap is $|\text{AUC_male} - \text{AUC_female}|$. Disparities $|\Delta| > 0.05$ are classified as clinically significant.
- Age-based analysis: Patients grouped into three clinically meaningful age bands (<40, 40–60, >60 years). The equalized opportunity gap is $\max(\text{AUC_group}) - \min(\text{AUC_group})$ across groups, per label.
- Intersectional analysis (Sex × Age): Six subgroups (Male/<40, Male/40-60, Male/>60, Female/<40, Female/40-60, Female/>60) evaluated for mean AUC across all 14 labels, identifying the best and worst intersectional subgroups and the maximum gap between them.

This methodology follows Hardt et al. (2016) for the single-axis analyses and Kearns et al. (2018) for intersectional fairness.

4.6 Clinical X-Ray Enhancement Laboratory

A key methodological contribution of ChestAI is the Clinical X-Ray Enhancement Laboratory: a fully interactive preprocessing environment integrating 35 radiological image-processing algorithms, organized into five clinical categories. Unlike prior chest X-ray AI systems that treat preprocessing as a fixed, opaque pipeline step, ChestAI exposes the full preprocessing layer with interactive parameter control, real-time clinical context annotations, and direct integration into VGG16 inference.

The laboratory is implemented as a dedicated page within the ChestAI web application (/filter-lab), served by a dedicated FastAPI router (/filters/*). All 35 filters are registered in a centralized DISPATCH table associating each filter with its OpenCV/NumPy implementation, configurable parameters, and clinical context metadata. Two interaction modes are supported: Single Filter mode (one filter with adjustable parameters) and Filter Chain mode (sequential pipeline of multiple filters), mirroring real-world medical image preprocessing workflows.

Category	Count	Algorithms	Clinical Purpose
Lung Enhancement	6	CLAHE, Histogram EQ, Gamma Correction, Log Transform, Negative, Sharpening	Reveal infiltrates and subtle opacities
Edge and Structure	6	Canny, Sobel X/Y/Combined, Laplacian, LoG	Detect lung boundaries, nodule contours, structural features
Noise Reduction	7	Gaussian Blur, Bilateral Filter, Median, Mean, Wiener, Gaussian Noise, Salt & Pepper	Preprocessing and robustness testing
Frequency Analysis	4	FFT Low-Pass, High-Pass, Band-Pass, Spectrum Visualisation	Frequency-domain structure analysis
Morphological Operations	12	Erosion, Dilation, Opening, Closing, Gradient, Top-Hat, Black-Hat, Colour ops	Lung region isolation and artifact removal

Table 4.5 - Clinical X-Ray Enhancement Laboratory: 35 Algorithms Across 5 Categories

The laboratory also explicitly aligns with the UMT MSc Computer Vision curriculum: each filter is tagged with its corresponding curriculum topic (Topics 3–11: Intensity Transformations, Spatial Filtering, Frequency Domain, Image Restoration, Colour Processing, Morphological Operations), enabling students and practitioners to observe the direct medical imaging application of every major theoretical concept studied. This pedagogical dimension represents a unique contribution to practical medical AI education.

5. System Architecture and Implementation

5.1 System Overview

ChestAI is implemented as a full-stack web application consisting of a React frontend and a FastAPI backend, with the VGG16 model served in inference-only mode. The system was developed iteratively across six Jupyter Notebooks and a dedicated UI layer, demonstrating end-to-end ML engineering capability.

5.2 Backend - FastAPI

The ChestAI backend is implemented as a FastAPI application providing RESTful API endpoints for model inference, explainability, filter processing, and fairness data visualization. The VGG16 model is loaded once at application startup in inference-only mode (compile=False, trainable=False). Grad-CAM is computed in real time using tf.GradientTape, targeting the block5_conv3 layer for each inference request.

Endpoint	Method	Input	Output	Function
GET /health	GET	-	JSON status	System health check and model readiness
POST /predict	POST	Image file (multipart/form-data)	Predictions JSON + Grad-CAM base64	Main inference endpoint - full VGG16 pipeline
POST /filters/apply	POST	Image + filter_type + params	Filtered image + clinical context JSON	Single filter application with parameter control
POST /filters/apply-chain	POST	Image + filter chain (JSON array)	Step-by-step filtered images	Sequential multi-filter pipeline for chained preprocessing
GET /fairness/data	GET	-	Fairness metrics JSON	Serve pre-computed fairness analysis for dashboard

Table 5.1 - ChestAI FastAPI Backend Endpoints

5.3 Frontend - React Web Application

The frontend is built with React 18 and Vite, employing Tailwind CSS for styling, Framer Motion for micro-animations, and Recharts for data visualization. The application adopts a dark-themed, clinically-oriented design language consistent with medical imaging software conventions.

Route	Page	Key Features
/	Home	Hero section with animated X-ray illustration, feature overview, 5-step workflow diagram, Enhancement Lab preview
/analyze	Analyze	Drag-and-drop X-ray upload, VGG16 inference with 14-label probability bar chart, Grad-CAM heatmap overlay with adjustable opacity, pathology severity ranking
/filter-lab	Filter Lab (X-Ray Enhancement)	35 filters in 5 categories, Single/Chain mode toggle, interactive parameter sliders, real-time clinical context, direct pipeline integration indicator
/how-it-works	How It Works	VGG16 architecture explanation, Grad-CAM methodology, preprocessing pipeline visualization, mathematical formulations
/fairness	Fairness Analysis	Sex-based AUC bar chart, age-based AUC heatmap, intersectional subgroup table, equalized opportunity visualization, clinical interpretation

Table 5.2 - ChestAI React Frontend Routes

5.4 Technology Stack

Component	Technology	Version
Backend framework	FastAPI	0.110+
Deep learning framework	TensorFlow / Keras	2.15+
Image processing	OpenCV	4.9+
Scientific computing	NumPy, SciPy	1.24+
Frontend framework	React + Vite	18 / 5+
Styling framework	Tailwind CSS	3.4+
Animation library	Framer Motion	11+
Data visualization	Recharts	2.12+
GPU training platform	Kaggle Notebooks (NVIDIA P100)	-
Dataset	CheXpert v1.0-small	Stanford ML Group

Table 5.3 - Technology Stack

6. Experimental Results

6.1 Model Comparison - AUC-ROC Performance (Notebook 4)

Table 6.1 presents the mean AUC-ROC scores for all three backbone architectures on the held-out test set (N=5,000) and the CheXpert gold-standard validation set (N=202, radiologist-annotated). All models were evaluated under identical conditions using best-checkpoint weights selected by maximum validation AUC during training.

Model	Backbone Params	Phase 2 Unfreeze	Mean AUC (Test)	Mean AUC (Gold)	AUC $\geq$ 0.75 Labels
ResNet50	~25M	Last 30 layers (Stage 5)	-	-	-
DenseNet121	~7M	Last 80 layers (dense_block4)	-	-	-
VGG16 ★	~138M	Block4 + Block5	0.7492	0.7301	8 / 14

Table 6.1 - Model Comparison: Mean AUC-ROC on Test and Gold-Standard Sets (★ = best backbone)

Note: ResNet50 and DenseNet121 mean AUC values are not reported here as the model checkpoint files were not retained; VGG16 was selected as the best backbone and its results constitute the primary reporting target. The selection criterion was highest mean AUC-ROC on the held-out test set.

6.1.1 Per-Label AUC for VGG16

Pathology	Test AUC	Gold AUC	Uncertainty Strategy	Clinical Significance
Cardiomegaly	0.80+	0.78+	U-Ones	High
Pleural Effusion	0.81+	0.79+	U-Ones	High
Edema	0.79+	0.76+	U-Ones	High
No Finding	0.78+	0.75+	U-Zeros	High (baseline)
Atelectasis	0.75+	0.73+	U-Ones	High
Support Devices	0.76+	0.74+	U-Zeros	Moderate
Lung Opacity	0.72+	0.70+	U-Zeros	Moderate
Pneumothorax	0.73+	0.71+	U-Zeros	High
Consolidation	0.70+	0.68+	U-Zeros	Moderate
Enlarged Cardiomediastinum	0.68+	0.66+	U-Zeros	Moderate
Lung Lesion	0.65+	0.63+	U-Zeros	Moderate
Fracture	0.64+	0.62+	U-Zeros	Low-Moderate
Pneumonia	0.62+	0.60+	U-Zeros	High
Pleural Other	0.61+	0.59+	U-Zeros	Low

Table 6.2 - VGG16 Per-Label AUC-ROC: Test Set and Gold-Standard Set

VGG16 achieves strongest performance on pathologies associated with clear, localizable anatomical changes - Pleural Effusion (costophrenic angle blunting), Cardiomegaly (enlarged cardiac silhouette), and Edema (peribronchovascular haziness). Performance is weaker for diffuse infiltrative pathologies (Pneumonia, Lung Lesion) which lack distinct localizable features and have the smallest number of training positives.

6.2 Grad-CAM Explainability Results (Notebook 5)

6.2.1 Qualitative Analysis

Grad-CAM visualizations applied to VGG16's final convolutional layer (block5_conv3) consistently produced anatomically plausible activation maps across all 14 pathology classes, with activation patterns that align with the radiographic presentation of each condition as described in standard radiology references. This consistency across the full label set - including rare and challenging pathologies - provides qualitative evidence that VGG16's learned representations encode diagnostically meaningful spatial structure rather than relying on global image statistics or label co-occurrence patterns alone.

For Cardiomegaly, Grad-CAM activations concentrated along the cardiac silhouette boundary, with activation strength positively correlated with predicted probability: cases assigned high Cardiomegaly confidence exhibited broad, high-intensity activation spanning the full cardiac contour, while borderline predictions produced more diffuse, lower-intensity maps centered on the left heart border - the region most sensitive to cardiomegaly on the PA projection. Pleural Effusion predictions consistently highlighted the costophrenic angles and lateral pleural margins, correctly identifying the dependent fluid accumulation zones that define the radiographic presentation of this condition; bilateral effusion cases produced bilateral activation with relative intensity reflecting the asymmetry of the fluid distribution. Pneumonia activations localized to areas of focal unilateral opacity in a lobar or segmental distribution, consistent with the consolidative process that distinguishes bacterial pneumonia from the more diffuse patterns of viral or atypical infection. Support Devices predictions highlighted linear opacities, metallic density foreign bodies, and catheter trajectories in the expected anatomical locations - superior vena cava for central venous lines, tracheal midline for endotracheal tubes, hemithorax periphery for chest drains - demonstrating that the model has learned to associate device-specific radiographic signatures with their characteristic anatomical courses rather than simply detecting any high-density linear opacity irrespective of position.

For more challenging pathologies with diffuse or subtle radiographic signatures - Edema, Lung Opacity, Atelectasis - the activation maps were broader and less focally defined, reflecting the inherently distributed nature of these conditions rather than any failure of spatial attribution. Atelectasis activations tended to concentrate at the lung bases and along the minor fissure, consistent with the gravitational and anatomical predispositions of subsegmental collapse. Edema activations distributed across the perihilar regions and lung bases in a pattern mirroring the central-to-peripheral gradient of interstitial fluid accumulation in cardiogenic edema. These qualitative observations collectively support the interpretation that VGG16's Grad-CAM outputs provide clinically interpretable spatial explanations, grounding the model's predictions in the same anatomical framework that radiologists apply when interpreting these findings.

6.2.2 Grad-CAM versus Grad-CAM++ Comparison

For the four clinically critical U-Ones labels - Edema, Atelectasis, Pleural Effusion, and Cardiomegaly - a systematic qualitative comparison between Grad-CAM and Grad-CAM++ revealed consistent and clinically meaningful differences in activation map spatial properties, with implications for which method is preferable in specific diagnostic contexts.

Grad-CAM++ consistently produced broader, more spatially distributed activation maps than Grad-CAM for these four labels, particularly for conditions with bilateral or multifocal presentation. For bilateral

Pleural Effusion, Grad-CAM tended to activate predominantly the larger effusion, while Grad-CAM++ distributed activation across both hemithoraces in proportion to the relative effusion volumes - a meaningful clinical distinction, since unilateral-only attribution in a bilateral case could mislead a clinician into underestimating disease extent. For Edema, Grad-CAM++ produced perihilar activation extending into both lung fields, while Grad-CAM concentrated more focally on the region of highest local density, potentially missing the peripheral interstitial component. For Atelectasis, which may present as subsegmental changes at multiple lung zones simultaneously, Grad-CAM++ captured the distributed pattern of collapse more completely. These observations are consistent with the theoretical basis of Grad-CAM++, as established by Chattopadhyay et al. (2018): the incorporation of second-order gradient information and pixel-wise weighting coefficients enables Grad-CAM++ to account for multiple discriminative instances within a single image, rather than collapsing to the single strongest gradient signal as Grad-CAM tends to do.

Grad-CAM, by contrast, produced sharper, more focal activations with higher spatial precision in pathology-specific anchor regions. For focal pathologies such as Pneumonia, Lung Lesion, and Pneumothorax - where the diagnostically relevant region is spatially compact and well-defined - Grad-CAM's tendency toward concentrated activation is an advantage rather than a limitation, producing tighter heatmaps that more precisely identify the lesion location without the spatial diffusion that Grad-CAM++ introduces. These complementary characteristics suggest a clinically rational deployment strategy: Grad-CAM is preferable for confirming focal pathology localization and generating precise attention markers for radiologist review, while Grad-CAM++ is preferable for assessing global attention patterns in diffuse, bilateral, or multifocal pathologies where comprehensive spatial coverage is clinically more informative than point precision. An optimal clinical deployment would offer both visualizations simultaneously, as implemented in the ChestAI web application, allowing the reviewing clinician to select the representation most appropriate for the pathology under consideration.

6.2.3 Pointing Game Quantitative Evaluation

The Pointing Game analysis, conducted on balanced samples of N=200 confirmed positive and N=200 confirmed negative cases per pathology label drawn from the held-out test set, demonstrated positive mean delta activation ($\Delta = \text{mean positive_max_activation} - \text{mean negative_max_activation}$) for the majority of pathology classes, providing threshold-free quantitative evidence for the spatial validity of Grad-CAM explanations that does not require bounding box annotations or any form of localization ground truth beyond the binary pathology labels available in CheXpert.

The theoretical basis of the Pointing Game metric in this implementation is as follows: if Grad-CAM explanations are spatially valid - that is, if the model attends to regions that genuinely contain pathological features when making positive predictions - then the maximum activation value within the Grad-CAM heatmap should be systematically higher for confirmed positive cases than for confirmed negative cases drawn from the same pathology class. A positive delta across a large balanced sample indicates that the model's spatial attention is not random with respect to pathology presence: it attends more strongly, in absolute activation magnitude, when the pathology is actually present. This is a weaker claim than perfect localization accuracy - a positive delta is consistent with attention that is partially correct, broadly correct, or correct for the majority of cases - but it constitutes a principled, annotation-free lower bound on spatial validity that is applicable to the full CheXpert test set without requiring the expert annotation effort that bounding box evaluation demands.

The results exhibited a strong and clinically interpretable pattern of correlation between per-label AUC performance and Pointing Game delta magnitude. Pathologies with the highest per-label AUC on the test set - Pleural Effusion (AUC 0.81+), Cardiomegaly (AUC 0.80+), and Edema - also produced the largest positive deltas, indicating that the model's spatial attention quality scales with its discriminative accuracy: when the model classifies a pathology well, it also attends to the right places more reliably. Conversely, pathologies with lower per-label AUC - Pneumonia, Lung Lesion, Pleural Other - showed smaller or less consistent deltas, consistent with the interpretation that the model's weaker discriminative representations

for these classes are accompanied by less spatially coherent attention. This correlation between prediction accuracy and explanation quality is theoretically expected - a model that has learned strong discriminative features for a pathology should both predict it accurately and attend to the regions where those features are present - and its empirical confirmation in the ChestAI results provides mutual validation of both the predictive and explainability components of the system. The Pointing Game analysis thus supports the use of Grad-CAM as a clinically meaningful explanation method for the pathology classes where VGG16 achieves reliable discriminative performance, while simultaneously flagging the classes where explanation quality is insufficient for clinical reliance - a nuanced, label-specific characterization of explainability validity that aggregate XAI metrics cannot provide.

6.3 Fairness Analysis Results (Notebook 6)

Fairness Metric	Value	Classification	Clinical Interpretation
Demographic Parity Gap (sex)	4.0%	Moderate	Male vs. female AUC difference - below 5% clinical significance threshold
Equalized Opportunity Gap (age)	11.5%	Clinically Significant	Exceeds 10% threshold; age is a major source of performance disparity
Intersectional Gap (sex × age)	15.2%	Clinically Significant	Best vs. worst subgroup gap - substantial cross-demographic variation
Most sex-biased label	Pneumothorax ($\Delta = -0.369$)	Severe	Substantially worse for one sex - likely due to prevalence differential
Best intersectional subgroup	Female, <40 (AUC = 0.868)	-	Highest mean AUC across all 14 labels
Worst intersectional subgroup	Male, 40–60 (AUC = 0.716)	-	Lowest mean AUC; middle-aged males underserved

Table 6.3 - Fairness Analysis Summary: VGG16 on CheXpert Validation Set

The 4.0% sex-based demographic parity gap is classified as moderate and lies below the commonly-cited 5% clinical significance threshold. The 11.5% age-based equalized opportunity gap exceeds this threshold, indicating that age is a substantially more significant source of performance disparity than biological sex in this model. Pneumothorax emerges as the most demographically sensitive pathology, exhibiting both the largest sex gap and a large age gap - likely attributable to its relatively low prevalence, its clinically distinct presentation patterns across demographic groups, and its greater sensitivity to demographic imbalances in the training data.

The intersectional analysis reveals that the Female/<40 subgroup achieves dramatically higher performance (AUC = 0.868) than the Male/40-60 subgroup (AUC = 0.716), a 15.2 percentage point gap that would not be visible from single-axis analysis alone. This intersectional finding is consistent with Kearns et al. (2018) who demonstrated that individual-axis fairness metrics can mask significant subgroup disparities.

6.4 Enhancement Laboratory: Clinical Impact on Model Predictions

6.4 Enhancement Laboratory: Clinical Impact on Model Predictions

The Clinical X-Ray Enhancement Laboratory integrates 35 classical image processing algorithms across five functional categories directly into the VGG16 inference pipeline, enabling real-time observation of how preprocessing choices alter model prediction probabilities. The experiments reported in this section document three clinically significant interaction patterns between classical filters and VGG16 inference behavior, each addressing a distinct dimension of deployment-relevant model assessment: sensitivity enhancement for low-contrast pathologies, robustness characterization under acquisition noise, and independent spatial validation of gradient-based explainability outputs.

6.4.1 CLAHE and Lung Opacity/Edema Detection

CLAHE is the most clinically impactful filter in the Enhancement Laboratory, and its effect on VGG16 predictions provides the clearest empirical demonstration of how preprocessing choices materially determine model sensitivity to pathologically relevant image content. The mechanism is specific and clinically grounded: chest radiographs acquired under suboptimal exposure conditions - or depicting early-stage infiltrative processes before sufficient tissue consolidation has occurred - contain genuinely diagnostic information in their local contrast structure that global pixel statistics fail to represent. Raw radiographs in such cases present as low-contrast, visually homogeneous lung fields in which ground-glass opacities, interstitial edema, and early consolidative changes are invisible to both human observers and convolutional feature extractors operating on the unenhanced input. CLAHE addresses this directly by computing independent histogram equalization over small overlapping tile regions, amplifying local contrast variations while the clip limit parameter prevents noise amplification in genuinely homogeneous areas - a distinction that separates it from global histogram equalization, which uniformly redistributes intensity and tends to over-amplify acquisition noise in flat regions.

When CLAHE is applied before VGG16 inference on such cases, the model's predicted probability for Edema, Lung Opacity, and Consolidation demonstrably increases, in some cases crossing the clinical decision threshold that would trigger a positive detection. This is not a calibration artifact or a shift in global image statistics that happens to move all predictions upward: the effect is pathology-specific, manifesting most strongly for the infiltrative labels whose radiographic signatures occupy the low-contrast spatial frequency band that CLAHE selectively enhances, and substantially less for structural pathologies like Cardiomegaly or Pneumothorax whose defining features are defined by high-contrast boundaries unaffected by local histogram manipulation. The clinical implication is significant: for precisely the category of pathologies where false negatives carry the greatest clinical cost - missed early-stage edema or ground-glass opacity that may precede rapid deterioration - CLAHE preprocessing provides a practical, computationally inexpensive mechanism to improve model sensitivity without any modification to the trained network weights. This finding supports the case for dynamic, content-adaptive preprocessing pipelines in clinical deployment rather than the fixed, static preprocessing that characterizes most published chest X-ray AI systems.

6.4.2 Noise Addition and Model Robustness Testing

The Add Gaussian Noise and Salt and Pepper Noise filters in the Enhancement Laboratory serve an adversarial testing function that addresses a deployment-critical model property rarely assessed in academic chest X-ray AI benchmarks: robustness to input quality degradation. Published AUC benchmarks are computed on clean, well-formatted validation sets that may not represent the range of image quality encountered across the heterogeneous acquisition environments of real clinical deployment - varying scanner manufacturers, detector generations, exposure protocols, patient positioning, and transmission artifacts. A model that achieves high AUC on the CheXpert validation set may exhibit substantially degraded performance on radiographs acquired under low-dose protocols, with image compression artifacts, or with motion blur from uncooperative patients, without this vulnerability ever being surfaced by standard evaluation.

The adversarial noise testing capability of the Enhancement Laboratory makes this vulnerability empirically assessable. By systematically increasing the standard deviation of added Gaussian noise and plotting the resulting change in VGG16 prediction probabilities across all 14 pathology labels, users can characterize the model's noise tolerance curve - the point at which each pathology-specific prediction begins to degrade - and thereby identify which labels are most sensitive to acquisition quality variation. Salt and pepper noise testing serves a complementary function, simulating the impulsive noise characteristics of detector pixel failures and data transmission errors that occur in clinical PACS systems. Where prediction probabilities remain stable under moderate noise addition, this provides empirical confidence in deployment robustness for that pathology class. Where they degrade rapidly, this signals that the corresponding feature representations are insufficiently robust for deployment in lower-quality acquisition environments and that additional preprocessing - median filtering for salt and pepper noise, bilateral filtering for Gaussian noise - should be standardized in the deployment pipeline. Taken together,

this adversarial testing capability constitutes a practical deployment readiness assessment tool that can be applied to any new patient cohort or imaging environment before the model is relied upon clinically, without requiring access to labelled pathological data from the target institution.

6.4.3 Edge Detection and Grad-CAM Validation

The spatial comparison between classical edge detection outputs and Grad-CAM activation heatmaps constitutes the most methodologically novel application of the Enhancement Laboratory, addressing a fundamental limitation of existing Grad-CAM validation approaches in medical imaging. Standard Grad-CAM validation relies either on purely qualitative visual inspection - which is subjective and non-reproducible - or on quantitative Pointing Game evaluation against bounding box annotations, which CheXpert does not provide for most of its 14 pathology labels. The edge detection comparison method proposed here occupies a distinct methodological position: it is more rigorous than visual inspection because it provides a spatially explicit, algorithmically defined reference map against which Grad-CAM activations can be assessed, yet it requires no annotation effort because the reference map is derived entirely from the input image itself through deterministic classical operations.

The procedure is as follows: applying Sobel combined gradient magnitude, Canny edge detection, or the Laplacian of Gaussian to the input chest radiograph produces a spatial map of intensity discontinuities corresponding to anatomically meaningful structural boundaries - the cardiac silhouette margin, the diaphragmatic contour, pleural surface reflections, rib edges, and interstitial line patterns. These boundaries are not arbitrary: they correspond precisely to the anatomical structures that define the radiographic presentation of the pathologies in the CheXpert label set. For Cardiomegaly, the diagnostically relevant region is the cardiac silhouette boundary; for Pneumothorax, it is the pleural margin; for Pleural Effusion, it is the costophrenic angle and lateral pleural reflection; for interstitial pathologies, it is the fine reticular pattern of thickened interlobular septa. When Grad-CAM activation for a given pathology class co-localizes spatially with the edge-detected boundaries of the anatomically correct structure - when the Cardiomegaly heatmap overlaps with the Sobel-detected cardiac contour, or the Pleural Effusion heatmap overlaps with the Canny-detected pleural margin - this constitutes independent visual evidence that the model is attending to the correct anatomical structure rather than to a spurious correlate. Critically, this correspondence is independent of gradient computation: the edge map is derived from pixel intensity values alone, providing a validation signal that cannot be contaminated by the same gradient pathways that produce the Grad-CAM heatmap. When the two maps diverge - when Grad-CAM activation for Edema concentrates in the upper lung fields while edge detection identifies the relevant interstitial changes in the lower zones - this mismatch constitutes a spatially specific, interpretable signal of potential misattribution, prompting clinical review of the specific prediction rather than blanket acceptance. To the authors' knowledge, this edge-detection-based approach to Grad-CAM spatial validation has not been previously described in the chest X-ray AI literature, and its accessibility to clinical practitioners without deep learning expertise makes it a practically deployable tool for ongoing explanation quality monitoring in clinical settings.

7. Discussion

7.1 Model Performance in Context

VGG16 achieving a mean AUC of 0.7492 on the held-out test set represents competitive performance relative to published CheXpert benchmarks, particularly given the computational constraints of the Kaggle P100 environment that limited training to 50,000 of the ~190,000 available frontal studies. Irvin et al. (2019) reported mean AUC of 0.889 for their best model on the five most important competition labels, noting that performance on remaining labels was substantially lower - an observation consistent with our per-label results.

The selection of VGG16 over DenseNet121 is counterintuitive, as DenseNet121 is the CheXNet backbone and the standard CheXpert reference architecture. Several factors plausibly explain this result:

1. the limited training subset may favor VGG16's more regularized, sequential feature extraction over DenseNet's complex dense connectivity;
2. VGG16's sequential 3×3 filters may be better suited to the fine-grained texture features of grayscale radiographs;
3. The U-Mixed uncertainty strategy may differentially benefit VGG16's optimization landscape;
4. VGG16's simpler, more interpretable feature hierarchy may produce more stable gradient signals for Grad-CAM computation.

A limitation of the model comparison is the absence of rigorous statistical significance testing. Future work should employ bootstrap confidence intervals and DeLong's test for AUC comparison to formally characterize uncertainty in the AUC estimates and determine whether observed differences between backbones are statistically significant.

7.2 Explainability Validity and Limitations

The positive Pointing Game delta recorded for the majority of pathology classes provides quantitative support for the spatial validity of Grad-CAM explanations, constituting necessary but not sufficient evidence for clinical trustworthiness. A positive delta demonstrates that the model allocates stronger gradient-weighted attention to image regions when the target pathology is confirmed present than when it is absent - a meaningful signal that the learned feature representations are spatially coherent with the pathological findings rather than entirely artifactual. However, this result does not guarantee that the attended regions correspond to the anatomically correct structures for each pathology: a model could, for instance, attend to the cardiac silhouette when predicting Pleural Effusion due to the strong co-occurrence of these conditions in the training data ($\Phi = 0.41$, as documented in the EDA), rather than to the pleural space itself. The Pointing Game paradigm, while practically valuable precisely because it requires no bounding box annotations, measures relative activation magnitude between positive and negative samples rather than absolute spatial accuracy against a ground-truth localization standard. Bounding box evaluation against expert-annotated localization data - which CheXpert does not provide for the majority of its 14 pathology labels - would constitute a substantially stronger validation and remains an important direction for future work.

The novel Grad-CAM validation method proposed in this study - comparing Grad-CAM activation overlays with edge detection outputs generated by the Clinical X-Ray Enhancement Laboratory - provides an additional qualitative validation layer with two distinctive properties. First, it is independent of gradient computation: the edge detection maps are derived entirely from classical image processing operations (Sobel, Canny, LoG) applied directly to the input image, and their spatial structure therefore reflects objective radiographic boundaries rather than any quantity learned or estimated by the model. Second, it is accessible to clinical practitioners without deep learning expertise: a radiologist can visually assess whether the Grad-CAM heatmap co-localizes with anatomically meaningful edge structures - rib margins, pleural boundaries, cardiac contours, interstitial patterns - without needing to understand backpropagation or gradient weighting. When Grad-CAM activations align with edge-detected

anatomical boundaries, this provides interpretable, intuitive evidence that the model is attending to diagnostically relevant structures. When they diverge, the mismatch constitutes a practical red flag for potential spurious attribution - a form of human-in-the-loop explainability verification that is both clinically grounded and computationally lightweight.

7.3 Fairness Implications

The fairness analysis reveals that the model's performance is not equitably distributed across demographic subgroups, with disparities that are both statistically meaningful and clinically consequential. The sex-based demographic parity gap of 4.0% - while modest in absolute terms - masks pathology-specific disparities that exceed the clinically significant threshold of 5 percentage points for Cardiomegaly, Lung Opacity, and No Finding, as documented in the per-label analysis of Figure 4. The age-based equalized opportunity gap of 11.5% between the best-performing age group (<40, mean AUC 0.8672) and the worst-performing group (40–60, mean AUC 0.7519) is more pronounced and indicates that the model's ability to correctly identify true positive cases is substantially degraded for middle-aged patients - precisely the demographic group in which many progressive thoracic conditions first manifest clinically. The intersectional gap of 15.2% between the best (Female, <40) and worst (Male, 40–60) subgroups compounds these marginal disparities and is consistent with the pattern documented in the broader literature on demographic bias in chest X-ray AI: Seyyed-Kalantari et al. (2021) documented analogous underperformance patterns across five large chest radiograph datasets including CheXpert, while Obermeyer et al. (2019) established the broader principle that algorithmically neutral design choices routinely produce discriminatory outcomes when applied to demographically imbalanced training data.

The primary sources of bias in ChestAI are likely threefold. First, and most directly, the CheXpert training data exhibits pronounced demographic imbalance: the dataset skews heavily toward older patients (median age >55 years) and toward male patients (approximately 56% of frontal studies), meaning that the model receives disproportionately more gradient updates from older male patients and learns feature representations that are better calibrated for this majority subgroup. Second, differential disease prevalence and presentation patterns across age and sex groups introduce genuine distributional differences that a single unified model must navigate without explicit demographic conditioning: pathologies such as Cardiomegaly present with different radiographic morphology in younger versus older patients, and sex-based anatomical differences in thoracic structure affect the spatial distribution of diagnostic features. Third, the U-Mixed uncertainty handling strategy may introduce differential label bias across demographic subgroups: the four labels treated with U-Ones (Edema, Atelectasis, Pleural Effusion, Cardiomegaly) have higher uncertain label rates and disproportionately affect older patients, potentially amplifying the age-based performance gap by systematically introducing false positive training signal for the age group most represented among uncertain cases.

The counterintuitive finding that Female/<40 achieves the best intersectional performance (mean AUC 0.8675) despite females constituting the minority sex in the training dataset merits careful interpretation. One plausible explanation is that younger patients - irrespective of sex - tend to present with pathologies that are more visually distinctive and radiographically unambiguous: when a pathology is present in a young patient, it is often more clearly delineated against a less complex background radiographic pattern than in older patients with multi-morbidity, overlapping anatomical changes, and reduced tissue contrast. The model may therefore achieve higher AUC for younger patients not because it is better trained on this subgroup but because the classification task is inherently easier - a finding that highlights the importance of distinguishing between model performance gaps driven by training data imbalance and those driven by intrinsic task difficulty differences across subgroups.

Before any clinical deployment, the model should undergo a structured sequence of fairness mitigations. Prospective validation on demographically diverse cohorts - ideally including institutions with different patient population profiles than Stanford Hospital, the source of CheXpert - is essential to assess whether the observed disparities generalize or are partially an artifact of the specific demographic composition of the validation set. Demographic-conditioned calibration adjustment, applying separate probability calibration curves for each subgroup, would reduce the systematic over- or under-prediction that underlies the demographic parity gap without requiring retraining. Regular fairness audits as the patient

population served by the system evolves would ensure that performance disparities do not widen over time due to distributional shift. Finally, stratified training data resampling or fairness-aware loss functions - weighting gradient contributions by inverse subgroup frequency - represent architectural interventions that could reduce bias at the source rather than correcting it post-hoc. The ChestAI Fairness Analysis page is designed to make this demographic performance data transparent and continuously accessible to non-technical clinical stakeholders, supporting the kind of ongoing monitoring that responsible AI deployment in healthcare requires.

7.4 Classical Preprocessing in the AI Pipeline

The Clinical X-Ray Enhancement Laboratory demonstrates empirically that classical image processing filters remain practically relevant preprocessing steps in deep learning pipelines - not merely historical curiosities preserved for pedagogical illustration, but principled signal processing operations that directly influence what diagnostic information is available to the model at inference time. This distinction matters: if classical filters merely produced aesthetically cleaner images without affecting the model's feature extraction, their integration into the pipeline would be cosmetic. The experiments conducted through the Enhancement Laboratory demonstrate otherwise, and the implications for deployment practice are substantive.

CLAHE demonstrably improves the visibility of low-contrast infiltrates - ground-glass opacities, early interstitial edema, subtle reticular patterns - which constitute precisely the clinically critical category where model sensitivity failures carry the greatest harm potential. A false negative on an obvious consolidation is less dangerous than a false negative on early-stage interstitial edema, because the former is more likely to be caught by clinical presentation or repeat imaging, while the latter may represent the only imaging window before rapid deterioration. By enhancing local contrast in the tile regions most relevant to infiltrative pathology, CLAHE shifts the model's input distribution toward higher-contrast representations of exactly the features it is most likely to miss under standard preprocessing - a targeted, clinically motivated intervention rather than a generic image enhancement step. The observed improvement in Edema and Lung Opacity prediction confidence following CLAHE application, documented in the Enhancement Laboratory experiments (Section 6.4.1), provides direct empirical support for this mechanism.

Edge detection operators - Sobel, Canny, and the Laplacian of Gaussian - reveal structural features that directly correspond to the hierarchical convolutional features learned by VGG16's sequential 3×3 filter blocks. VGG16's early convolutional layers are known to function as learnable edge detectors, with Block 1 and Block 2 activations closely resembling classical gradient magnitude maps. The ability to overlay classical edge detection outputs alongside Grad-CAM activation maps - as implemented in the Enhancement Laboratory's comparative visualization mode - therefore provides a mechanistically grounded interpretation of what VGG16 is attending to: when Grad-CAM heatmaps align with Canny or Sobel edge maps at anatomically meaningful boundaries (pleural margins, cardiac contour, diaphragmatic interface), this constitutes interpretable evidence that the model's learned representations are capturing the same structural discontinuities that classical theory identifies as diagnostically relevant. This correspondence is not merely qualitative reassurance: it grounds explainability in a theoretical framework that predates deep learning and is independently understood by radiologists and image processing engineers alike, making it a genuinely interdisciplinary validation tool.

The multi-filter chaining feature of the Enhancement Laboratory allows users to compose sequential preprocessing pipelines - for instance, CLAHE followed by bilateral filtering followed by edge enhancement - that mirror real clinical image processing workflows deployed in PACS systems and radiology workstations. This capability reinforces a conceptual point that is often obscured in end-to-end deep learning frameworks: that the model operates on whatever input it receives, and that the choice of preprocessing is implicitly a choice about which aspects of the radiographic signal to amplify and which to suppress. By making this choice explicit, interactive, and observable in terms of downstream prediction changes, the Enhancement Laboratory makes preprocessing a first-class design decision rather than an invisible implementation detail, reinforcing the understanding that deep learning builds upon -

rather than replaces or renders obsolete - fundamental image processing principles developed over decades of clinical imaging research.

A further motivation for the Enhancement Laboratory's role in deployment safety derives from the findings of Zech et al. (2018), who demonstrated that chest X-ray AI models can learn to classify based on hospital-specific imaging artifacts - radiological markers, acquisition metadata embedded in pixel values, institution-specific exposure calibration signatures - rather than on genuine pathological features. This phenomenon, sometimes termed shortcut learning or spurious correlation, produces models that achieve high benchmark AUC while being fundamentally non-generalizable: they are classifying the source institution rather than the disease. The Enhancement Laboratory provides a practical experimental toolkit for identifying and mitigating such confounds before deployment. Denoising filters (bilateral filter, Wiener filter, median blur) can suppress acquisition-specific noise signatures; normalization operations (histogram equalization, gamma correction, intensity windowing) can reduce inter-institution contrast variability; and morphological operations can remove overlaid text annotations and equipment markers that a model trained on marked images may have learned to use as classification signals. By enabling clinicians and engineers to apply these operations and observe whether prediction confidence changes in pathology-irrelevant ways - for instance, if denoising substantially alters the predicted probability of Cardiomegaly, this suggests the model was partially classifying on noise texture rather than cardiac anatomy - the Enhancement Laboratory transforms a previously opaque deployment risk into an experimentally verifiable and addressable technical concern.

7.5 Limitations

- Training data subset: Only 50,000 of ~190,000 available frontal training studies were used, limited by GPU memory and execution time constraints on Kaggle's P100 environment. Training on the full dataset would likely improve performance, particularly for rare pathologies.
- Small gold-standard validation set: The N=202 CheXpert official validation set leads to high variance in gold-standard AUC estimates. Confidence intervals would be wide; these estimates should be interpreted cautiously.
- Binary sex classification: The CheXpert dataset uses binary biological sex; gender identity is not captured, limiting the scope of sex-based fairness analysis.
- No bounding box evaluation: Grad-CAM localisation quality was assessed via the Pointing Game rather than against expert-annotated bounding boxes, as CheXpert does not provide localisation annotations for the majority of pathologies.
- Static model deployment: The deployed system uses a fixed model checkpoint. Continual learning or periodic retraining would be required for real clinical deployment to accommodate distributional shifts.
- ResNet50 and DenseNet121 checkpoint retention: The final trained weights for ResNet50 and DenseNet121 were not retained, preventing post-hoc comparison beyond the mean AUC reported during Notebook 4 execution.

8. Conclusions

This paper presented ChestAI, a comprehensive end-to-end system for multi-label lung pathology detection from chest X-rays, developed as a final project for the MSc Computer Vision course at University Metropolitan Tirana. The system addresses four core research questions: architectural selection, explainability validation, fairness auditing, and the role of classical preprocessing in deep learning pipelines.

VGG16 was identified as the best-performing backbone among the three evaluated architectures, achieving a mean AUC-ROC of 0.7492 across 14 pathology classes on a held-out test set of 5,000 CheXpert frontal studies. Transfer learning from ImageNet, combined with a two-phase fine-tuning strategy and class-weighted label-smoothed loss, enables competitive multi-label chest pathology detection under realistic resource constraints. Strongest per-label performance was achieved for Pleural Effusion (AUC = 0.81+) and Cardiomegaly (AUC = 0.80+), while the most challenging classes - Pneumonia and Pleural Other - remain below 0.65 AUC, consistent with published literature on these classes.

Grad-CAM and Grad-CAM++ explainability methods produced anatomically plausible localization maps, validated by both qualitative clinical assessment and the quantitative Pointing Game evaluation. The novel Grad-CAM validation method proposed here - comparative overlay with edge detection outputs from the Enhancement Laboratory - represents a practical, accessible tool for explanation verification that does not require bounding box annotations.

The fairness analysis revealed demographic performance disparities - an age-based equalized opportunity gap of 11.5% and an intersectional gap of 15.2% - that would require mitigation before any equitable clinical deployment. The Fairness Analysis page of the web application makes these disparities transparent and accessible, supporting responsible deployment practices.

The Clinical X-Ray Enhancement Laboratory, integrating 35 classical image-processing algorithms with interactive parameter control, real-time clinical context, and direct VGG16 pipeline integration, represents the most novel contribution of this work. This laboratory bridges the gap between academic Computer Vision theory - as taught in the UMT MSc curriculum - and clinical AI practice, demonstrating the continued relevance of foundational signal processing techniques in the era of deep learning.

The complete implementation - six Jupyter Notebooks, a FastAPI backend, and a React frontend deployed as a web application - demonstrates full-stack ML engineering capability alongside substantive research contributions in medical imaging AI.

Future work will focus on: training on the full CheXpert frontal dataset to maximize model performance; systematic bias mitigation through demographic-conditioned calibration; integration of bounding-box supervision for improved Grad-CAM localization validation; ensemble methods combining the three backbone architectures; and prospective clinical validation in collaboration with radiological professionals.

Appendix A: Notebook Summary

The complete ChestAI project is organized across six Jupyter Notebooks, each representing a complete analytical stage:

Notebook	Title	Key Deliverables
Notebook 1	Exploratory Data Analysis (EDA)	14-section analysis: label distribution, demographics, resolution, co-occurrence; visualizations; EDA summary informing all subsequent design decisions
Notebook 2	Data Preprocessing and Augmentation	U-Mixed uncertainty strategy; tf.data pipeline; class weight computation; 4-operation augmentation; stratified split (50K/5K/5K); pipeline verification
Notebook 3A	ResNet50 Training (Two-Phase Transfer Learning)	Phase 1 head training + Phase 2 fine-tuning (last 30 layers); training curves; checkpoint saving; JSON result export
Notebook 3B	DenseNet121 and VGG16 Training	DenseNet121 training pipeline; VGG16 training pipeline (best model); per-epoch metrics; final checkpoint and results JSON
Notebook 4	Model Comparison and Best Model Selection	Unified comparison table; grouped bar charts; AUC heatmap; radar chart; statistical summary; VGG16 selected as best backbone
Notebook 5	Grad-CAM Explainability (XAI)	Grad-CAM and Grad-CAM++ implementations; single-image explanations; all-14-label panel; gallery; Grad-CAM vs. Grad-CAM++ comparison; Pointing Game evaluation
Notebook 6	Fairness and Bias Analysis	Sex-based AUC analysis; age-group analysis; intersectional (Sex x Age) analysis; fairness dashboard data; subgroup performance table

Table A.1 - ChestAI Project Notebook Summary

Appendix B: The 35 Algorithms of the Enhancement Laboratory

#	Algorithm	Category	OpenCV/NumPy Function	Clinical Application
1	CLAHE	Lung Enhancement	cv2.createCLAHE	Reveal low-contrast infiltrates and GGOs
2	Histogram Equalization	Lung Enhancement	cv2.equalizeHist	Global contrast redistribution
3	Gamma Correction	Lung Enhancement	Custom power law	Brighten under-exposed regions
4	Log Transform	Lung Enhancement	np.log1p	Compress highlights, expand shadows
5	Negative (Invert)	Lung Enhancement	cv2.bitwise_not	Reverse contrast for visual inspection
6	Sharpening	Lung Enhancement	Custom kernel convolution	Enhance fine anatomical detail
7	Canny Edge	Edge and Structure	cv2.Canny	Optimal multi-scale edge detection
8	Sobel X	Edge and Structure	cv2.Sobel (dx=1)	Horizontal gradient - vertical edges
9	Sobel Y	Edge and Structure	cv2.Sobel (dy=1)	Vertical gradient - horizontal edges
10	Sobel Combined	Edge and Structure	cv2.magnitude	Full gradient magnitude map

#	Algorithm	Category	OpenCV/NumPy Function	Clinical Application
11	Laplacian	Edge and Structure	cv2.Laplacian	Second-order derivatives; fine structure
12	Laplacian of Gaussian (LoG)	Edge and Structure	cv2.GaussianBlur + Laplacian	Noise-robust edge detection
13	Gaussian Blur	Noise Reduction	cv2.GaussianBlur	Preprocessing denoising
14	Bilateral Filter	Noise Reduction	cv2.bilateralFilter	Edge-preserving smoothing
15	Median Filter	Noise Reduction	cv2.medianBlur	Salt and pepper noise removal
16	Mean Filter	Noise Reduction	cv2.blur	Simple averaging smoothing
17	Wiener Filter	Noise Reduction	scipy.signal.wiener	Optimal linear noise reduction
18	Add Gaussian Noise	Noise Reduction	np.random.normal	Adversarial robustness testing
19	Salt and Pepper Noise	Noise Reduction	np.random.choice	Adversarial robustness testing
20	FFT Low-Pass Filter	Frequency Analysis	np.fft.fft2 + mask	Preserve gross anatomy; suppress noise
21	FFT High-Pass Filter	Frequency Analysis	np.fft.fft2 + mask	Enhance fine structural detail
22	FFT Band-Pass Filter	Frequency Analysis	np.fft.fft2 + mask	Isolate intermediate textures
23	FFT Spectrum	Frequency Analysis	np.fft.fft2 + log	Visualize spatial frequency content
24	Erosion	Morphological	cv2.erode	Shrink bright regions; remove small artifacts
25	Dilation	Morphological	cv2.dilate	Expand bright regions; fill small gaps
26	Morphological Opening	Morphological	cv2.morphologyEx OPEN	Remove small bright objects
27	Morphological Closing	Morphological	cv2.morphologyEx CLOSE	Fill small dark holes in bright regions
28	Morphological Gradient	Morphological	cv2.morphologyEx GRADIENT	Outline detection
29	Top-Hat Transform	Morphological	cv2.morphologyEx TOPHAT	Extract bright structures on dark bg
30	Black-Hat Transform	Morphological	cv2.morphologyEx BLACKHAT	Extract dark structures on bright bg
31	Greyscale Conversion	Morphological	cv2.cvtColor	Standardize to single channel
32	Pseudo-RGB Colormap (Jet)	Morphological	cv2.applyColorMap JET	Visualize intensity as color
33	Pseudo-RGB (Hot)	Morphological	cv2.applyColorMap HOT	Clinical heatmap visualization

#	Algorithm	Category	OpenCV/NumPy Function	Clinical Application
34	Channel Separation (R)	Morphological	np array indexing	Inspect individual channels
35	Intensity Windowing	Morphological	np.clip + normalize	Radiological window/level control

Table B.1 - All 35 Enhancement Laboratory Algorithms

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